

Capstone Project Introduction

ECON 490

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Overview

In these slides, we'll talk about:

- Requirements for the capstone project
- Basic components of your projects
- Tips for your proposals and common data analysis pitfalls

Key Components of Your Capstone Papers

Core components of your capstone projects:

1. Clear research question and motivation
2. Data analysis

We'll complete several intermediate assignments throughout the semester

- First step is the Research and Data Proposal
- Capstone Lab Activity 1: guided walkthrough of a common CPS data set we work through together as a class
- Capstone Lab Activity 2: repeat that same pipeline using your own capstone data
- Capstone Outline will include draft data analysis and presentations

Proposal/Outline assignments have detailed templates for you to use/submit

Capstone Project Grade Breakdown

The capstone project is worth **45%** of your final grade

<i>Assignment</i>	<i>Due Date</i>	<i>% of Final Grade</i>
Research Proposal	Wednesday, 10/7	5%
Capstone Lab Activity 1	Wednesday, 10/14	5%
Capstone Lab Activity 2	Wednesday, 10/21	5%
Outline Presentation	Wednesday, 11/4	2.5%
Outline	Friday, 11/6	7.5%
Final Presentation	Weeks 14/15 (12/2, 12/9)	10%
Final Paper	Friday, 12/18	10%

Final Deliverables

All capstone output should be:

- Written in your own words
- Neatly-formatted/presentable (no R screenshots unless specifically requested)

Final Papers: Roughly 8-10 pages

- Including data analysis output and works cited
- We will discuss writing and paper requirements later in semester

Presentations

- *Final Presentations:* ~15 minutes
- *Outline Presentations:* 5-10 minutes

Capstone Research and Data Proposal

First capstone assignment is due Week 7 (Wednesday, 10/7)

Steps to complete this assignment:

- 1) Decide if you want to work alone or with a partner
- 2) Identify a clear research question
- 3) Identify the data you'll use to answer your research question
- 4) Complete the proposal worksheet on Canvas

NOTE: *Before moving forward with your project, you will need approval on the topic and data source(s) – I may ask you to submit a revised proposal.*

Working Backwards

Trying to find a research question can be daunting

To make things easier, try *working backwards*

- Start thinking about the general topics you're interested in – this could be labor markets, housing, education, poverty, etc.
- Then ask, “What kind of data analysis can I do?” related to that topic

Your data analysis requires (1) data and (2) an analysis strategy

- Start by identifying and exploring the data set that best fits your interest
- Then, think about (2) possible analysis strategies
- Finally, return to research question and orient around your analysis

General Suggestions on Proposals

Key first step—find and download data (ideally multiple data sets)

- Load data in R, read documentation (!), run simple regressions
- Ask yourself, “Does this data look like something I can work with?”

Your first draft of a research question will likely be too superficial

- Remember, the goal is to write an 8+ page paper
- As part of this, you’ll spend 2-3 pages talking about your data analysis output

If your research question is something like, “What’s the correlation between GDP and unemployment?” you’re not going to have much to say!

Selection of Research Questions from Prior Semesters

How do recessions impact household food expenditures?

How do the returns to graduate education vary across demographic groups?

How did the Covid-19 pandemic impact household transportation choices?

What factors drive variation in home prices across counties in California?

How much of NBA player compensation can be explained by future expected performance (compared to past performance or prior accolades)?

Thinking Descriptively

How to go beyond, “What’s the correlation between X and Y?”

- For ECON 340, running 1-2 regressions is generally sufficient for a paper
- In 490, goal is to concretely answer a question/say something interesting

In general, think about *exploring variation* in either:

1. The level of one economic variable as a function of other variables
2. A relationship between economic variables

In both cases, you can explore variation across time, places, economic factors, etc.

Data Analysis Pitfalls – *Macro Time Series Data*

General rule of thumb: make sure you have data for multiple groups

- In a lot of data sets/research contexts, this happens naturally
- People in different states, players on different NBA teams, etc.

Avoid data for just a single place/company/person over time

- Lack of groups = lack of natural comparisons to make via regression
- Severely limits flexibility in your analysis = hard to write a 10-page paper

Want to do a macro topic? Look up data for multiple countries, or states in US

Data Analysis Pitfalls – *Ecological Fallacy*

A common 490 topic that I've decided to prohibit: *"What is the correlation between state economic conditions and crime?"*

Two problems:

- 1) Correlations are generally a bit weak
- 2) Correlations are very easy to misinterpret

Ecological fallacy = making inferences about individuals based on group-level data

- Asking, "How does losing my job affect my propensity to commit crime" is very different than asking, "How do recessions impact crime?"

Data Analysis Pitfalls – *Getting Too Fancy*

In general, OLS should be fully sufficient for most projects

- If you have a background/prior course experience with ML/time series metrics/etc., you are welcome to apply these tools!
- However, don't let AI push you into using techniques you haven't studied

Using what feels like a fancy method doesn't make your answer any better if the question isn't well phrased or the data isn't well-suited to your analysis

ChatGPT loves logit/probit regression and variance inflation factors

- Avoid these tools unless you 1) know why you're using them and 2) know how to correctly interpret their output